

🐼 3D Coordinates Extension — Build the Panda (Advanced)

Topic: 3D Panda (x, y, z) · **Course:** 3D Coordinates · **Level:** Advanced (Extension, after Lesson 6) ·

Time: about 45 minutes

Build a **standing panda** in 3D — a white body and head with black ears, black eye patches, and two black legs. Every block needs **three** numbers (**x, y, z**): x **across**, y **up**, z **deeper** (forward).

● Stand on your **home spot** (red block) every run. Move your feet, move your panda.

🎨 White parts = **white concrete**. Black parts = **black wool**.

1 · Predict 🧠

Read each set. Write your prediction **and the reason**.

```
place black block at (2, 3, 1)
place black block at (2, 4, 1)
place black block at (2, 5, 1)
```

Does this leg go across, up, or deeper? Why?


```
place white block at (2, 8, 1)
place white block at (2, 8, 2)
place white block at (2, 8, 3)
```

Across, up, or deeper? Why?


```
place white block at (2, 5, 1)
place white block at (5, 5, 1)
place white block at (2, 8, 1)
place white block at (5, 8, 1)
```

Where do these four blocks sit, and what shape do the corners mark? Why?


```
place white block at (2, 7, 1)
place black block at (2, 7, 1)
```

Same (x, y, z). White, black, or both? Why?

2 · Spot the Bug 🐛

Fix each one. Then say why it was wrong.

Bug A — This ear should sit one step **deeper**, at (2, 9, 2). It has only two numbers, so it lands flat on the front.

```
place black block at (2, 9)
```

Fixed code:

Why wrong? Why does your fix work? (2–3 sentences)

Bug B — The eye should be **up high on the front**: (5, 7, 1). The last two numbers are swapped.

```
place black block at (5, 1, 7)
```

Fixed code:

Why wrong? Why does your fix work? (2–3 sentences)

Bug C (*the tricky one*) — The code runs with no error, but the panda's eye ended up on the **side** of the head instead of the front. The white face is built at $z = 1$, but this eye was placed at $z = 2$ — one block too deep.

```
place white block at (2, 7, 1)
place black block at (2, 7, 2)
```

Fix it so the eye is on the front face, and explain what "one block too deep" means. (2–3 sentences)

3 · Show Your Work 📷 🎥

Walk around the 🐼 panda in the world. Then plan and build a **full panda** on your home spot, then add one change of your own.

Plan your coordinates first. The head is a square; the ears go up; the eyes sit on the front; the legs go down.

White head ($z = 1$): fill a square from (2, 5, 1) to (5, 8, 1).

Black ears (up, bigger y): (2, 9, 1) and (5, 9, 1).

Black eyes (front): (2, 7, 1) and (5, 7, 1).

Black legs (down, smaller y): fill (2, 3, 1) to (2, 4, 1) and (5, 3, 1) to (5, 4, 1).

Add depth: build the white head, ears, and legs again at $z = 2$ and $z = 3$. At $z = 2$ and $z = 3$, place plain white where the eyes were — the face shows on the **front only**.

Then a MODIFY challenge: change one thing your own way — a new colour, a bigger head, arms, or a bamboo stalk beside the panda.

💡 Tip: the eyes only show on the **front**. The sides and back are plain white.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera at the panda. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your **one** video to teacher on KakaoTalk.